

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE**

Winter Examination – 2022

Course: S. Y. B. Tech. Branch : Chemical Engineering Semester : III

Subject Code & Name: Fluid Flow Operations BTCHC 302

Max Marks: 60

Date:

Duration: 3hr

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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(Level/CO) Marks

Q. 1 Solve any Two of the following. 2x6
=12

- A) Derive an expression for hydrostatic equilibrium in a stationery mass of an incompressible fluid having cross section area 'S'. 2/1
- B) Air flows through the pipeline across which manometer is connected. Manometer is partially filled with water having density 1000 Kg/ m³. Level difference in the manometer is 19.2 cm. If the downside pressure, $P_b = 1.5$ atm, calculate the upstream pressure P_a .
Density of air = 1.2 kg/m³ 3/1
- C) State Bernoulli equation without friction. Explain corrections made to this equation for its practical use. 2/1

Q. 2 Solve any Two of the following. 2x6
=12

- A) Derive the Hagen Poiseuolli Equation and prove that $f = 16 / (NRe)$, for laminar flow. 2/2
- B) A pump draws a solution of density 1720 Kg/m³ from a storage tank of large diameter through a pipe of 75mm I.D. The average velocity in suction line is 1.25 m/sec. The pump discharges through 50 mm I.D. pipe. The end 3/2

of discharge pipe is at 40 m above the liquid level in the storage tank. Friction loss in entire piping amounts to 132.435 J/Kg & efficiency of the pump is 65% Calculate,

- A) Power required by pump.
- B) Pressure required to develop by pump.

- C) Whole milk at 293 K having density 1030 Kg/m³ and viscosity 2.12 cp is flowing at the rate 0.605 Kg/sec in a glass pipe having diameter 63.5 mm . 3/2
- i) Calculate Reynolds number. Is this turbulent flow.
 - ii) Calculate flow rate needed in Kg/ sec for a Reynolds number of 2100.

Q. 3 Solve Any Two of the following. 2x12=12

- A) Explain the flow of incompressible fluid through sudden expansion and derive equation for head loss h_{fe} . 2/3
- B) Define friction factor and with sketch and explain friction factor chart i.e. Moody's diagram. 2/3
- C) i) What is Drag? Define drag coefficient. 3/3
- ii) A bed of ion exchange beds 2.43 m high is backwashed with water to remove dirt . The specific gravity of particles is 1.25 and average size is 1.1 mm . Calculate minimum fluidization velocity.
Data : Minimum Porosity = 0.4; Sphericity =1.0

Q. 4 Solve any Two of the following. 2x6=12

- A) What are Positive displacement and Centrifugal pumps? Explain with sketch Reciprocating pump. 2/4
- B) It is planned to transfer molasses having viscosity 550 cp and density 1500Kg/m³ through 18 cm diameter steel pipe at a rate 900Kg/min. The line is to be 3.5 Km long . The delivery end is at 25m height than intake. The line contains 10 bends, 4 Elbows, 2 gate valves. Calculate total friction loss due to skin friction and fittings and calculate pump work if efficiency of pump is 70% 3/3

Data: for one elbow $k_f = 0.65$
 for one bend $k_f = 0.35$
 for one gate valve $k_f = 0.17$

- C) What is NPSH? How is it calculated? Why NPSH (Available) should be greater than NPSH (Required)? 3/3
- Q. 5 Solve any Two of the following. 2x12=12
- A) Differentiate between variable head meter and variable area meter. 2/5
 Explain with sketch Orifice meter.
- B) A pipe with 7.5 cm I.D. is carrying H_2SO_4 in which a venturimeter is fitted having throat dia 3.6 cm. The differential manometer filled with mercury connected across meter reads 36 cm. Calculate mass flow rate of acid through pipe. 3/5
 $C_v = 0.98$, $\rho_{acid} = 1310 \text{ kg/m}^3$.
- C) State Reynolds number, Power number and Froude number in agitation. Discuss different types of impellers used in agitation. 2/5